

API Testing with Insomnia and Swagger

Capstone 3

Version 7.1 Y

Import Insomnia Collections

There are 2 collections within this `Insomnia.yaml` file

`capstone-insomnia_collections.yaml`

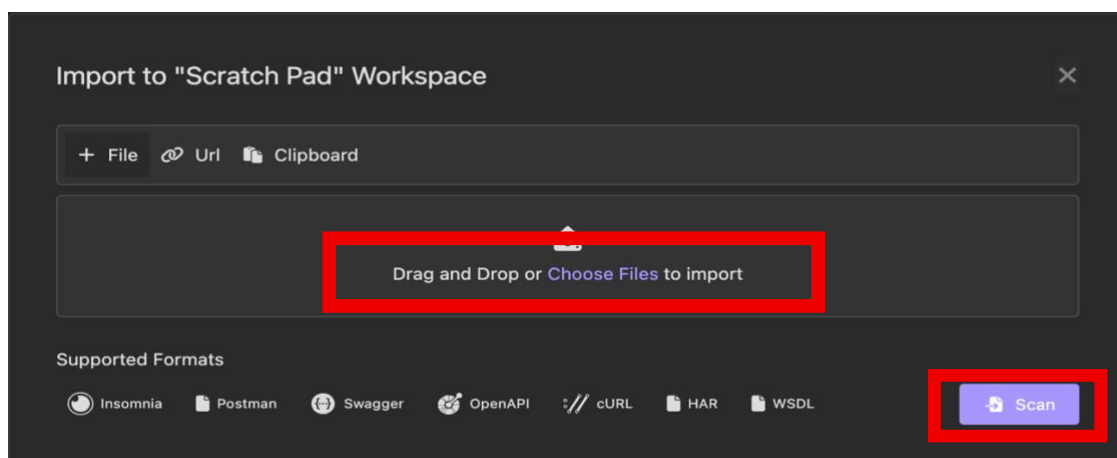
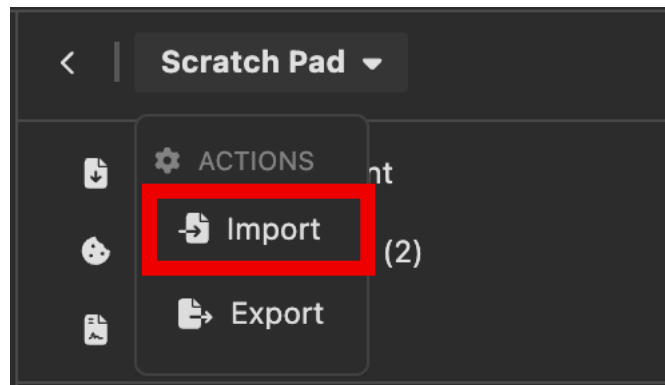
As the names indicate, the first is the collection of Insomnia REST requests for the required features in the capstone.

The second collection is a set of methods and tests for optional features and functionality in the capstone.

Import the Collections

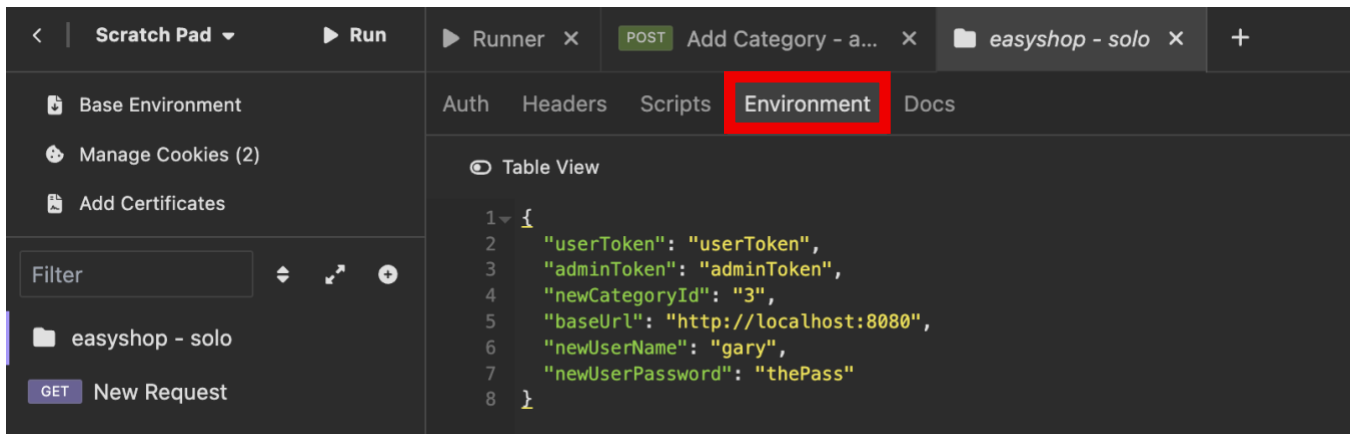
After creating the workspace, Insomnia will open the collections in the Scratch Pad. Click Import to launch the import tool.

Navigate to the folder with the insomnia json files and import both files.



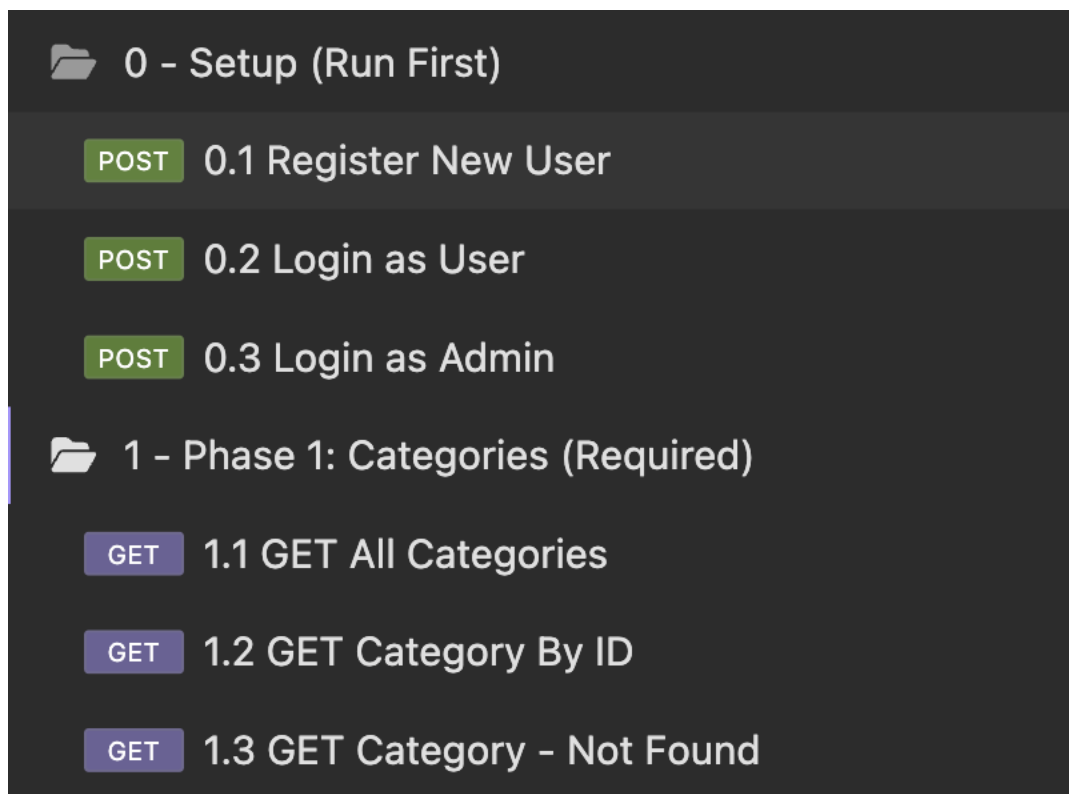
Testing with Insomnia

Insomnia allow us to create environment variables and use them to automate the testing process. Select the easyshop collection and click Environment to see the variables for this collection.

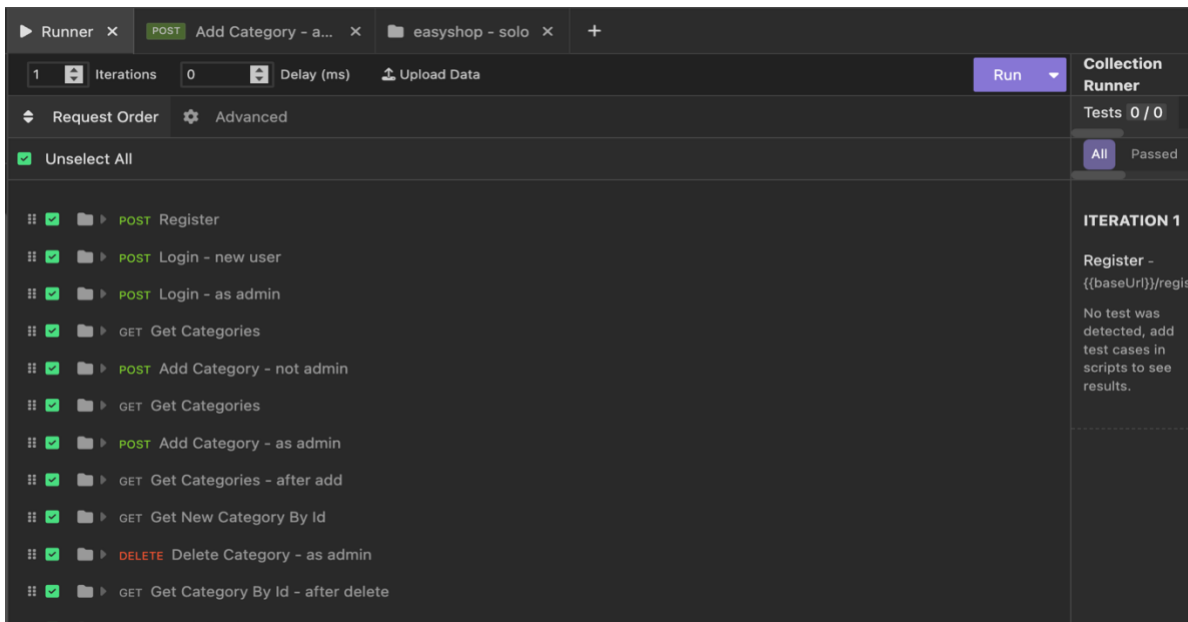


Running the Tests

In the 0 – Setup (Run First) collection, click the arrow button on the right and choose Run Folder to open the test suite. This is the first of multiple collections, labeled based on the phase of work you are in.

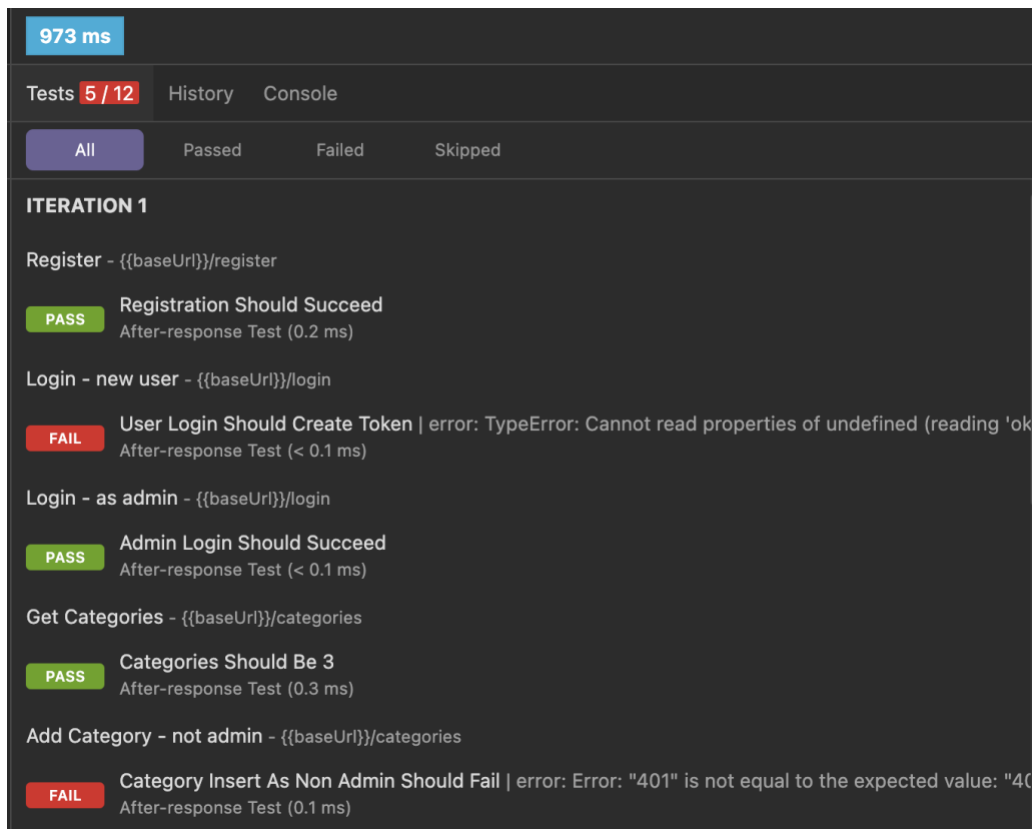


In the test suite you can choose which tests you want to run. If you want to test all the endpoints, click Select All



Tip: When Registering a user, usernames are typically unique and can't be created twice

Before you run the tests ensure that your capstone API projects is running in IntelliJ. Insomnia will run through all of the tests, and like Unit Tests, will indicate if the tests succeed or fail.



Testing with Swagger

What Is Swagger UI?

Swagger UI is a browser-based interface built into the application that lets you explore and test your API without any extra tools. It generates a visual reference of every available endpoint automatically based on the API's configuration.

Accessing the Interface

After starting the application, open your browser and navigate to:

`http://localhost:8080/swagger-ui/index.html`

Endpoints are grouped by phase. Any endpoint marked @todo is part of your implementation assignment.

Products Browse, search, and manage products		
GET	/products	Search / list products
POST	/products	Add product — Admin only
GET	/products/{id}	Get product by ID
PUT	/products/{id}	Update product — Admin only
DELETE	/products/{id}	Delete product — Admin only

Authenticating Your Requests

[Try it out](#)

Most endpoints require authentication. To set it up:

1. Send a request to POST /login and copy the token from the response.
2. Click the Authorize button (lock icon, top right).
3. Paste your token and click Authorize.

All subsequent requests will include your token automatically.

Request body **required**

```
{
  "username": "johndoe",
  "password": "p@ssw0rd",
  "confirmPassword": "p@ssw0rd",
}
```

Note: Refreshing the page clears your token.
You will need to re-authorize.